F7AB - Actuarial and Financial Mathematics
Tutorial 10 – Term structure of interest rates

Assume that time is measured in years. Let

$$\begin{aligned} f_{t,n} &= n\text{-year forward rate for an investment made at time } t \text{ for } n \text{ years.} \\ y_n &= \text{spot rate from time 0 to time } n. \\ c_n &= \text{par yield from time 0 to time } n. \\ P(n) &= \text{price at time 0 of a zero coupon bond that matures at time } n. \end{aligned}$$

Question 1 You are given that

$$\begin{aligned} f_{0,1} &= 6.0\% \text{ p.a. effective,} \\ f_{0,2} &= 6.5\% \text{ p.a. effective, and} \\ f_{1,2} &= 6.6\% \text{ p.a. effective.} \end{aligned}$$

Determine the 3-year par yield.

Question 2 The following n-year spot rates p.a. effective were observed at time $t = 0$.

n	1	2	3	4	5	6
y_n	4%	5%	6%	7%	7.5%	8%

1. Define what is meant by an n-year spot rate.
2. Calculate $f_{3,2}$.
3. Using the above n-year spot rates, calculate the 6-year par yield at time $t = 0$.

Question 3 You are given

$$\begin{aligned} f_{0,1} &= 6.0\% \text{ p.a. effective.} \\ f_{0,2} &= 6.5\% \text{ p.a. effective.} \\ f_{1,2} &= 6.6\% \text{ p.a. effective.} \end{aligned}$$

1. Calculate the 2-year spot rate.
2. Calculate the 3-year spot rate.

Question 4 You are given the yield to redemption on a n-year bond with a 6% coupon payable annually, for $n = 1, 2, 3$ as follows.

n	1	2	3
Yield to redemption (p.a. effective)	6%	6.3%	6.6%

All bonds are redeemed at par (i.e. the redemption proceeds equal 1) and are exactly one year from the next coupon payment.

1. Calculate y_1, y_2, and y_3 assuming no arbitrage.
2. Calculate $f_{0,1}$, $f_{1,1}$, and $f_{2,1}$ assuming no arbitrage.
3. Explain why the forward rates above increase more rapidly with term than the spot rates.

Question 5 You are given the following information about par yields (p.a. effective).

n	1	2	3	4	5	6	7	8
c_n	0.060	0.057	0.056	0.058	0.062	0.064	0.064	0.064

Calculate the values of $P(n)$ and y_n for $n = 1, 2, \ldots, 8$.

Question 6 Denote by $F(0, t, T)$ the forward rate (p.a. effective) at time 0 that applies between times t and T.

You are given the following information about par yields.

n	1	2	3
c_n	0.05263	0.05405	0.03261

1. Calculate the values of $P(n)$ for $n = 1, 2, 3$.
2. Hence derive the forward rates $F(0, 1, 2)$, $F(0, 1, 3)$ and $F(0, 2, 3)$.
3. Comment on the arbitrage possibilities presented by the specified values of the par-yield curve.

Answers: Q1. 6.395% p.a.; Q2.2. 9.790% p.a.; Q2.3. 7.708% p.a.; Q3.1. 6.5% p.a.; Q3.2. 6.400% p.a.; Q4.1. $y_1 = 6\%$ p.a, $y_2 = 6.309\%$ p.a, $y_3 = 6.625\%$ p.a.; Q4.2. $f_{0,1} = 6\%$ p.a., $f_{1,1} = 6.619\%$ p.a, $f_{2,1} = 7.259\%$ p.a.; Q5. Some sample solutions: $P(1) = 0.943396$ and $y_1 = 0.06$, $P(2) = 0.895200$ and $y_2 = 0.056915$, $P(5) = 0.738112$ and $y_5 = 0.062614$, $P(8) = 0.605757$ and $y_8 = 0.064664$; Q6.1. $P(1) = 0.950001$, $P(2) = 0.900007$, $P(3) = 0.909996$; Q6.2. $F(0, 1, 2) = 5.555\%$, $F(0, 1, 3) = 2.174\%$, $F(0, 2, 3) = -1.098\%$.

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Solutions to Tutorial 10 – Term structure of interest rates

Solution 1 The par yield c_3 satisfies

$$\frac{c_3}{(1+f_{0,1})} + \frac{c_3}{(1+f_{0,2})^2} + \frac{1+c_3}{(1+f_{0,1})(1+f_{1,2})^2} = 1.$$

Substituting for $f_{0,1}$, $f_{0,2}$ and $f_{1,2}$ and rearranging, we get

$$2.655250\, c_3 = 0.169806 \qquad \Rightarrow \qquad c_3 = 6.395\% \text{ p.a. effective.}$$

Solution 2

1. The n-year spot rate is the constant effective annual rate of interest we obtain in the market by buying 1 unit of an n-year zero-coupon bond at time 0 and holding it until time n.

2. $$(1+f_{3,2})^2 = \frac{(1+y_5)^5}{(1+y_3)^3} = \frac{(1.075)^5}{(1.06)^3} = 1.205382$$

   Hence $f_{3,2} = (1.205382)^{0.5} - 1 = 0.09790 = 9.790\%$ p.a. effective.

3. The 6-year par yield c_6 satisfies

   $$1 = c_6 \times \left((1+y_1)^{-1} + (1+y_2)^{-2} + \cdots + (1+y_6)^{-6}\right) + 1 \times (1+y_6)^{-6}.$$

   Substituting for the values of y_n,

   $$1 = c_6 \times \left((1.04)^{-1} + (1.05)^{-2} + (1.06)^{-3} + (1.07)^{-4} + (1.075)^{-5} + (1.08)^{-6}\right) + (1.08)^{-6}.$$

   Solving for c_6, we find

   $$c_6 = (1 - (1.08)^{-6})/4.797811 = 0.07708 = 7.708\% \text{ p.a. effective.}$$

Solution 3

1. $y_2 = f_{0,2} = 6.5\%$ p.a. effective.

2. The 3-year spot rate y_3 satisfies

   $$(1+y_3)^3 = (1+f_{0,1})(1+f_{1,2})^2 = (1.06)(1.066)^2 = 1.204537.$$

   Hence $y_3 = 6.400\%$ p.a. effective.

Solution 4

1. Denote by $P_n^{0.06}$ the price at $t = 0$ of the n-year bond with a 6% coupon payable annually for each £1 of nominal.

   In an arbitrage-free market, these coupon-bond prices and the spot rates are related by the following equation

   $$P_n^{0.06} = 0.06 \times \left((1+y_1)^{-1} + (1+y_2)^{-2} + \cdots + (1+y_n)^{-n})\right) + (1+y_n)^{-n}. \qquad (1)$$

   We wish to find y_1, y_2 and y_3. Using the yields to redemption, we can calculate $P_1^{0.06}$, $P_2^{0.06}$ and $P_3^{0.06}$ as follows.

   $$P_1^{0.06} = 1.06/1.06 = 1.$$
   $$P_2^{0.06} = 0.06\left((1.063)^{-1} + (1.063)^{-2}\right) + (1.063)^{-2} = 0.994523.$$
   $$P_3^{0.06} = 0.06\left((1.066)^{-1} + (1.066)^{-2} + (1.066)^{-3}\right) + (1.066)^{-3} = 0.984138.$$

   Now we can solve (1) iteratively. Setting $n = 1$, we find

   $$y_1 = 0.06.$$

   Setting $n = 2$ in (1),

   $$\frac{1.06}{(1+y_2)^2} = P_2^{0.06} - \frac{0.06}{1+y_1}$$
   $$\Rightarrow (1+y_2)^2 = 1.06\left(P_2^{0.06} - \frac{0.06}{1+y_1}\right)^{-1}$$
   $$\Rightarrow (1+y_2)^2 = 1.06\left(0.994523 - \frac{0.06}{1.06}\right)^{-1}$$
   $$\Rightarrow y_2 = \left(1.06 \times (0.937919)^{-1}\right)^{1/2} - 1 = 0.0630906.$$

   Setting $n = 3$ in (1),

   $$\frac{1.06}{(1+y_3)^3} = P_3^{0.06} - \frac{0.06}{1+y_1} - \frac{0.06}{(1+y_2)^2}$$
   $$\Rightarrow (1+y_3)^3 = 1.06\left(P_3^{0.06} - \frac{0.06}{1+y_1} - \frac{0.06}{(1+y_2)^2}\right)^{-1}$$
   $$\Rightarrow (1+y_3)^3 = 1.06\left(0.984138 - \frac{0.06}{1.06} - \frac{0.06}{(1.0630906)^2}\right)^{-1}$$
   $$\Rightarrow y_3 = \left(1.06 \times (0.874444)^{-1}\right)^{1/3} - 1 = 0.0662473.$$

   i.e. $y_1 = 6\%$ p.a. effective, $y_2 = 6.309\%$ p.a. effective and $y_3 = 6.625\%$ p.a. effective.

2. Clearly, $f_{0,1} = y_1 = 0.06$.

   $$f_{1,1} = \frac{(1+y_2)^2}{(1+y_1)} - 1 = 0.0661902.$$

   $$f_{2,1} = \frac{(1+y_3)^3}{(1+y_2)^2} - 1 = 0.0725888.$$

   i.e. $f_{0,1} = 6\%$ p.a. effective, $f_{1,1} = 6.619\%$ p.a. effective and $f_{2,1} = 7.259\%$ p.a. effective.

3. The n-year spot rate y_n is related to the one-year forward rates $\{f_{t,1} : t = 0, 1, \ldots\}$ by the following formula:

$$(1 + y_n)^n = (1 + f_{0,1}) \times (1 + f_{1,1}) \times \cdots \times (1 + f_{n-1,1}).$$

i.e. $(1 + y_n)$ is the geometric mean of $(1 + f_{0,1}), \ldots, (1 + f_{n-1,1})$.

Thus if $f_{0,1}, \ldots, f_{n-1,1}$ are increasing then y_n also increases but at a slower rate.

Note: the forward rates $\{f_{t,1} : t = 0, 1, \ldots\}$ may decrease.

Solution 5 The key formula is

$$c_n \left(P(1) + P(2) + \cdots + P(n)\right) + P(n) = 1, \tag{2}$$

which we can solve iteratively to find $P(1), P(2), \ldots, P(8)$. Using these calculated values, we can calculate $y_1, y_2, \ldots, y_8$ via the relationship

$$P(n) = (1 + y_n)^{-n}.$$

To iterate (2), re-arrange it as

$$P(n) = \frac{1 - c_n \left(P(1) + P(2) + \cdots + P(n-1)\right)}{1 + c_n}.$$

Then as $P(1) = (1 + c_1)^{-1} = (1.060)^{-1} = 0.943396$, we obtain the values in the table.

n	c_n	$P(n)$	y_n
1	0.060	0.943396	0.060
2	0.057	0.895200	0.056915
3	0.056	0.849468	0.055887
4	0.058	0.797819	0.058093
5	0.062	0.738112	0.062614
6	0.064	0.685775	0.064886
7	0.064	0.644525	0.064759
8	0.064	0.605757	0.064664

Solution 6

1. As in the previous question, calculate the zero-coupon bond prices:

n	c_n	$P(n)$
1	0.05263	0.950001
2	0.05405	0.900007
3	0.03261	0.909996

2. Then as $(1 + F(0, n, m))^{m-n} = P(n)/P(m)$,

$$F(0,1,2) = \frac{P(1)}{P(2)} - 1 = 0.055548 = 5.555\% \text{ p.a.}$$

$$F(0,1,3) = \left(\frac{P(1)}{P(3)}\right)^{0.5} - 1 = 0.021744 = 2.174\% \text{ p.a.}$$

$$F(0,2,3) = \frac{P(2)}{P(3)} - 1 = -0.010977 = -1.098\% \text{ p.a.}$$

3. Negative money interest rates present us with an arbitrage opportunity, since we could borrow money at the negative rate and put the hard cash in a safe (or under the bed) rather than put it in a bank account at the same negative rate.